PARKTOWN BOYS' HIGH SCHOOL

Department of Mathematics

Preliminary Examination Paper 2

3 September 2024

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner	Mrs S Rowell
Moderator	Ms H Gouws

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 10 questions, 12 pages and a formula sheet. You may detach the formula sheet for your use.
   Please ensure that your question paper is complete.

2. Answer ALL the questions in the answer books provided. Additional space is provided at the end of the answer book.

3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.

4. Answers only will NOT necessarily be awarded full marks.

5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.

6. If necessary, round off answers to TWO decimal places, unless stated otherwise.

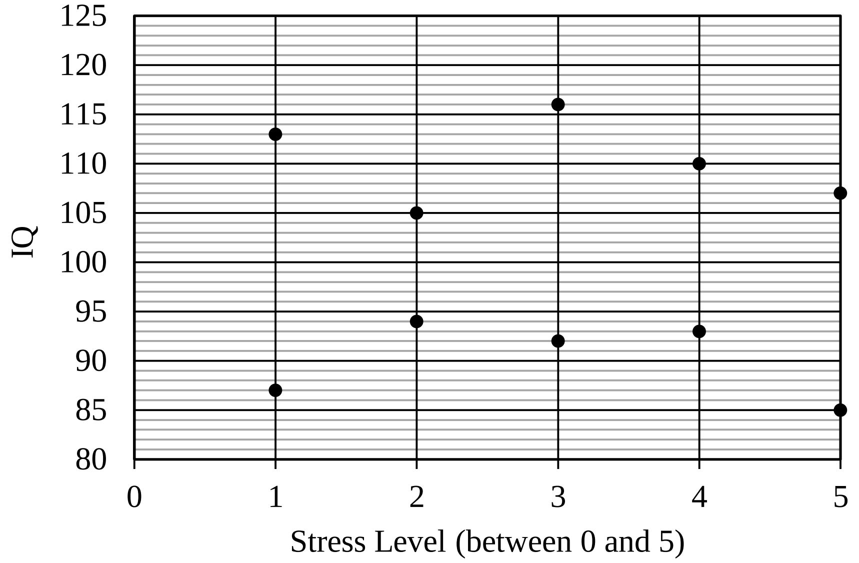
7. Diagrams are NOT necessarily drawn to scale.

8. Use the mark allocation to guide the length of your solution.

9. Write neatly and legibly.

QUESTION 1

The scatter plot below shows the stress levels of ten matric learners when compared to their intelligence quotient (IQ). The stress levels were observed between levels 0 to 5, 0 being not stressed at all and 5 being extremely stressed.



The same ten learners reported the number of hours of sleep they get in a week and the results were as shown below:

Stress Level (x)	1	3	2	5	4	4	1	2	3	5
Weekly number of hours of sleep (y)	56	45	52	33	35	36	58	50	46	37

1.1 Calculate the

1.1.1 Mean number of hours of sleep. (2)

1.1.2 Standard deviation of the number of hours of sleep. (1)

1.2 The learners who had fewer hours of sleep than one standard deviation below The mean was invited to a counselling session. How many learners were invited? (2)

1.3 Determine the equation of the least squares regression line for the data given in the table. (2)

1.4 Predict the number of hours of sleep a learner will get if he has a stress level of 4. (2)

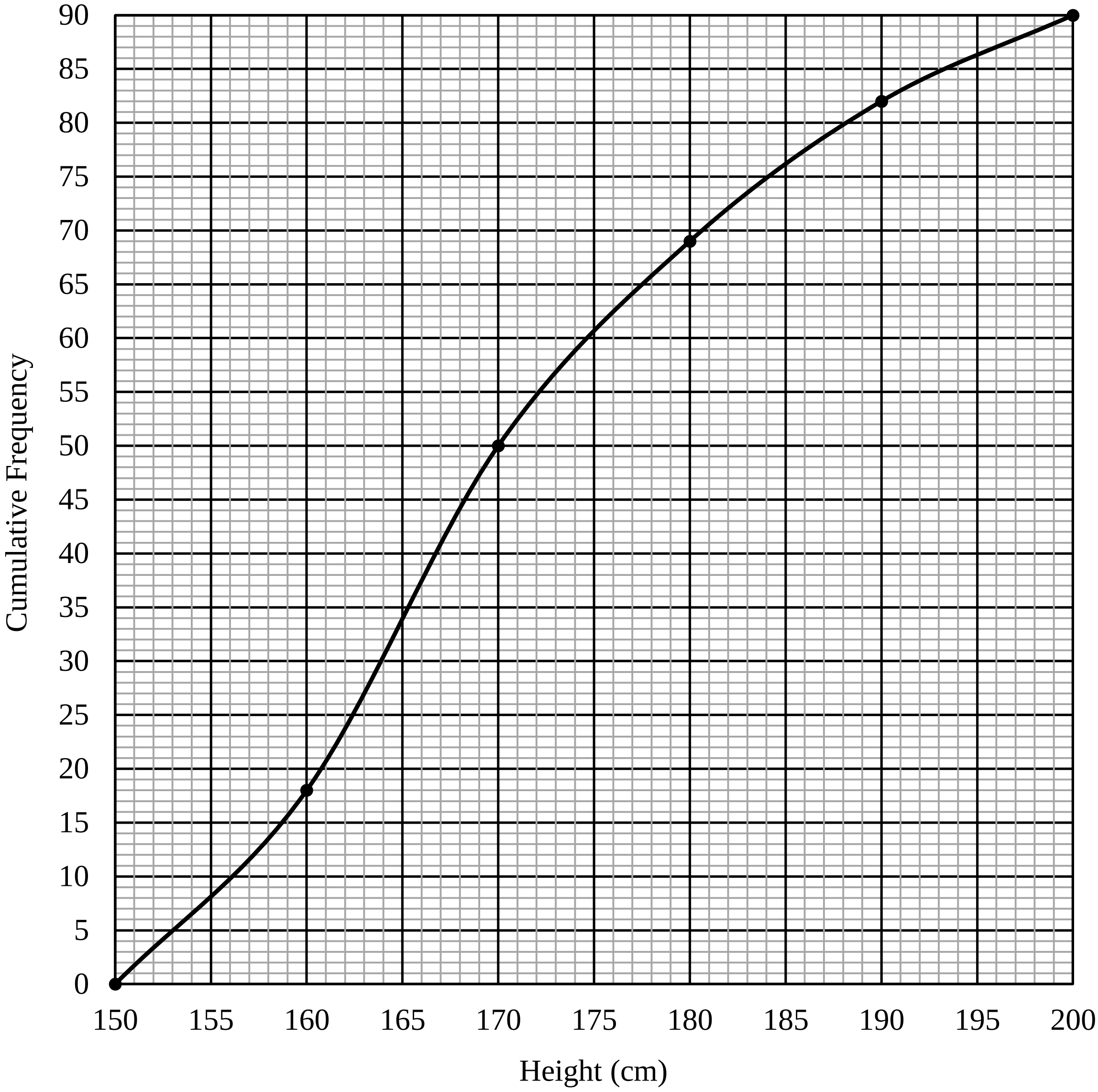
1.5 Calculate the correlation coefficient and comment on the relationship between stress levels and number of hours of sleep. (3)

1.6 Is IQ a good indicator of a learner's stress level? Motivate your answer, including a suitable calculation. (3)

[15]

QUESTION 2

The results of a study on the heights of a group of matric Parktown Boys' learners are shown in the cumulative frequency graph (ogive) below.

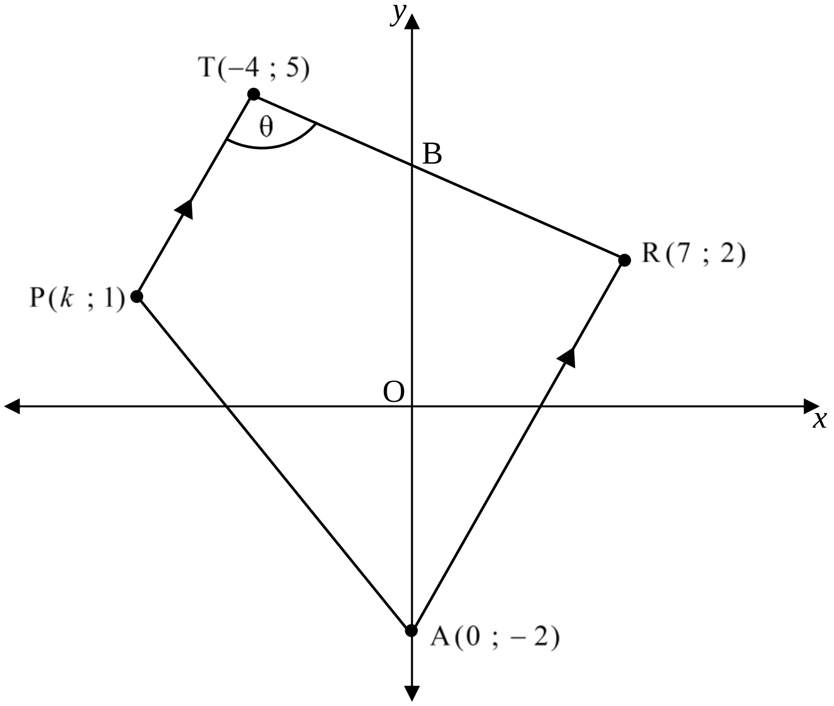


2.1	How many matrics participated in the study?	(1)
2.2	What is the modal class?	(1)
2.3	Use the graph to estimate the	
	2.3.1 median.	(2)
	2.3.2 interquartile range.	(3)
2.4	What percentage of the matric learners are taller than 185 cm?	(3)

[10]

QUESTION 3

In the diagram below, TRAP is a trapezium with vertices $T(-4\ ;\ 5)$, $R(7\ ;\ 2)$, $A(0\ ;\ -2)$ and $P(k\ ;\ 1)$. $TP \parallel RA$ and $P\hat{T}R = \theta$.

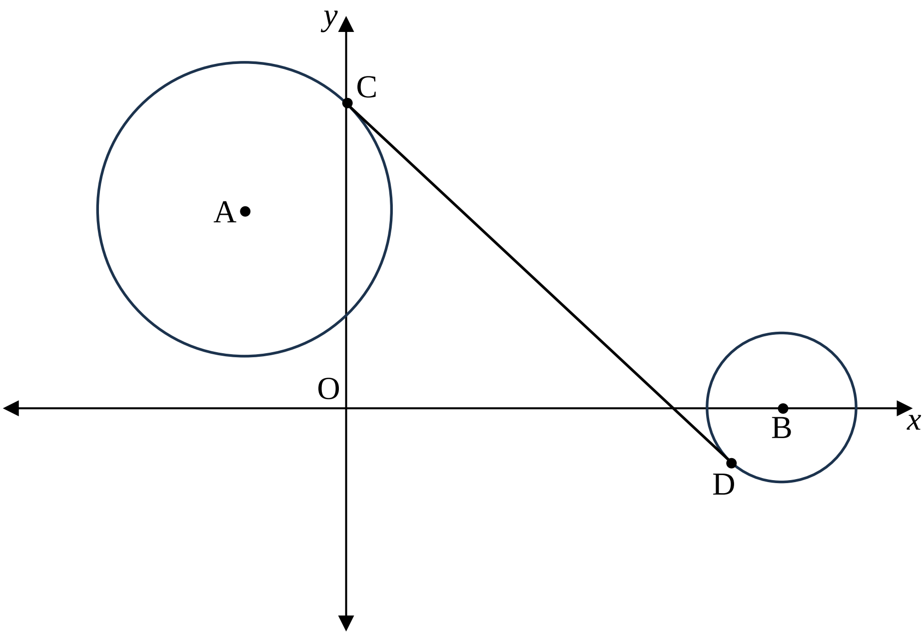


3.1	Determine the equation of TR.	(4)
3.2	If TRAM is a parallelogram, determine the coordinates of M.	(2)
3.3	Calculate the area of ΔABR.	(3)
3.4	Determine the value of *k*.	(4)
3.5	Calculate the size of θ.	(4)

[17]

QUESTION 4

In the diagram below, the circle with equation $x^2 + y^2 + 8x - 12y + 27 = 0$ has a centre at A and cuts the y-axis at C. The circle with centre B has the equation $(x-10)^2 + y^2 = t$. CD is a tangent to circle A at C and circle B at D.



4.1 Determine the coordinates of

4.1.1 A (3)

4.1.2 C (2)

4.2 Determine the equation of CD. (4)

4.3 If AC : BD = 5 : 2, determine the value of t. (3)

4.4 Determine the equation of the circle BCD. (6)

[18]

QUESTION 5

5.1 If $17\sin x = -8$, $x + y = 270°$ and y is an acute angle, determine, without the use of a calculator and with the aid of a diagram the value of

5.1.1 $\cos y$. (5)

5.1.2 $\tan x + \tan y$. (3)

5.2 If $\sin 24° \cos 15° = p$ and $\cos 24° \sin 15° = q$, determine the following in terms of p and q:

5.2.1 $\sin 9°$ (2)

5.2.2 $\cos 18°$ (2)

5.3 Simplify the following to a single trigonometric ratio without the use of a calculator:

$$1 + \frac{\sin(90° + \theta) \cdot \cos(180° + \theta)}{\sin(\theta - 60°) \cdot \sin\theta + \cos\theta \cdot \cos(\theta - 60°)}$$

(5)

5.4 Consider the identity $\dfrac{2\tan x}{1 + \tan^2 x} = \sin 2x$.

5.4.1 Prove the identity. (5)

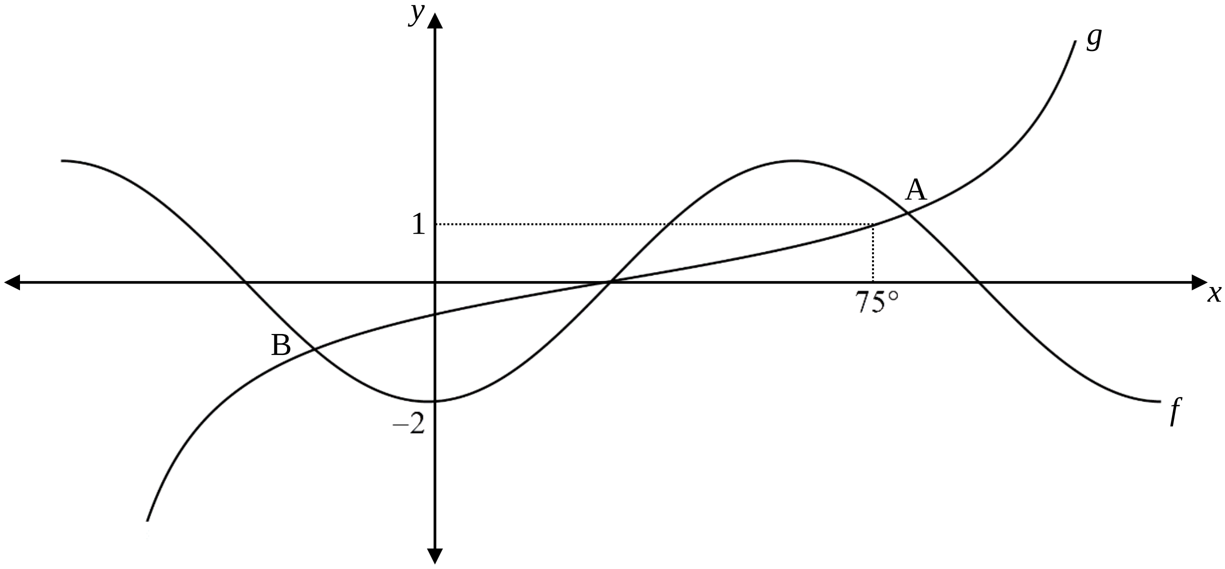
5.4.2 For which values of $x \in [-270° ; 270°]$ is this identity undefined? (2)

5.5 Determine the general solution of $\sin x \cos x = -0,24$. (4)

[28]

QUESTION 6

In the diagram below, the graphs of $f(x) = a\cos bx$ and $g(x) = \tan(x - c)$ have been sketched for $x \in [-60° ; 120°]$. The graph of g passes through the point $(75° ; 1)$. A and B are points of intersection of f and g.



6.1 Determine the values of *a*, *b* and *c*. (3)

6.2 What is the period of f? (1)

6.3 Write down the equation of the asymptote of $g(x + 25°)$ for the given interval. (1)

6.4 Given that the one general solution of $f(x) = g(x)$ is $x = 77{,}5° + k \cdot 96°$; $k \in \mathbb{Z}$. Determine the coordinates of B. (2)

6.5 Determine the value(s) of *x* for which:

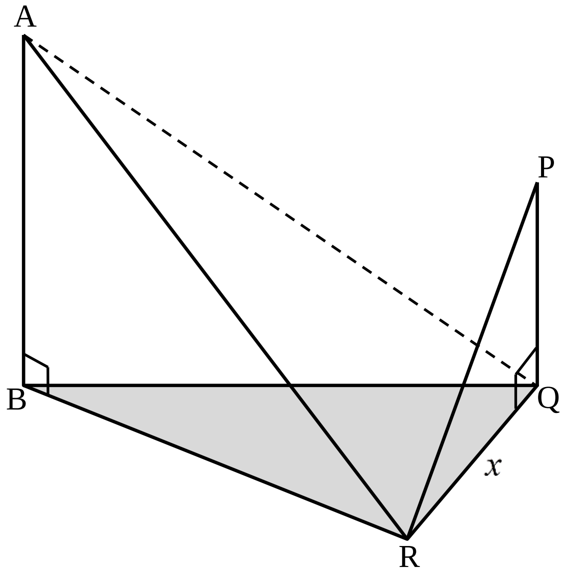
6.5.1 $f(x) = 2$ (1)

6.5.2 $f(x) \cdot g(x) \leq 0$ (3)

[11]

QUESTION 7

In the diagram below, R, Q and B lie on the horizontal plane. PQ and AB are vertical towers. From R, the angles of elevation to P and A are θ and 2θ respectively.
$QR = x$, $A\hat{Q}R = 90° + \theta$ and $Q\hat{A}R = \theta$.



7.1 Prove that $AB = 2x\cos^2\theta$. (5)

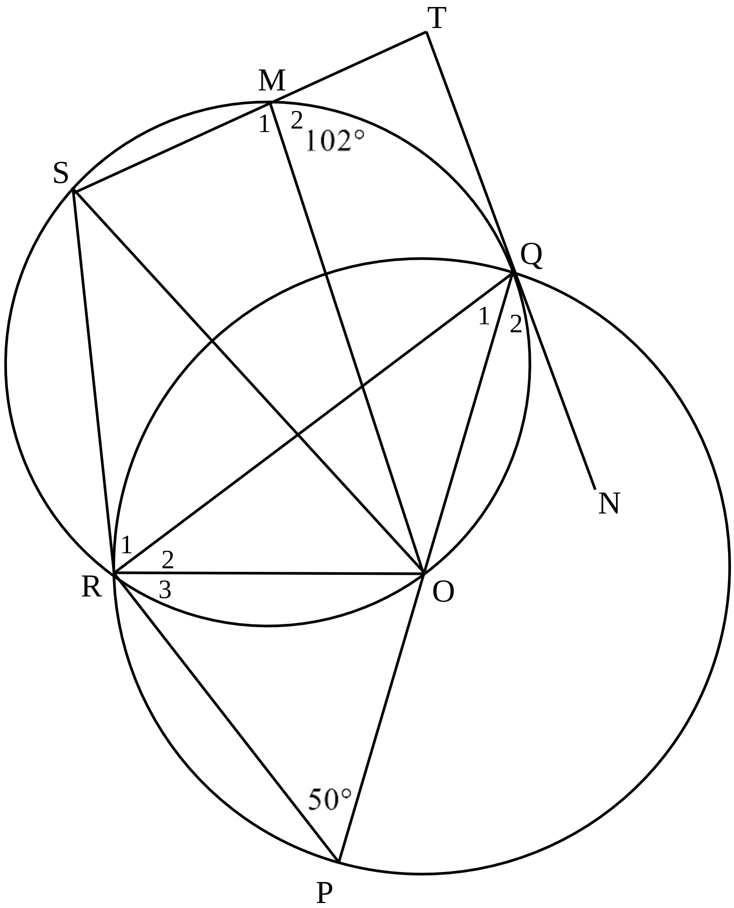
7.2 Hence, prove that $\frac{AB}{PQ} = \frac{2\cos^2\theta}{\tan\theta}$. (2)

7.3 If $\theta = 30°$, calculate $\frac{AB}{PQ}$, without the use of a calculator. (3)

[10]

QUESTION 8

In the diagram below, POQ is a diameter of circle PRQ. A smaller circle intersects the larger circle at R and Q. TN is a tangent to the smaller circle at Q. SM is produced to T.
$\hat{P} = 50°$ and $\hat{M}_2 = 102°$.



Determine, with reasons, the size of:

8.1 $\hat{Q}_1$ (3)

8.2 $R\hat{O}Q$ (2)

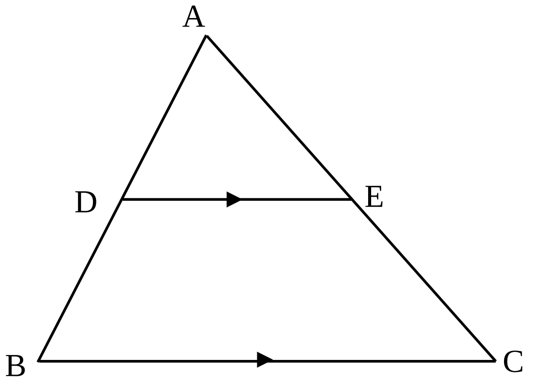
8.3 $\hat{Q}_2$ (2)

8.4 $\hat{R}_1$ (2)

[9]

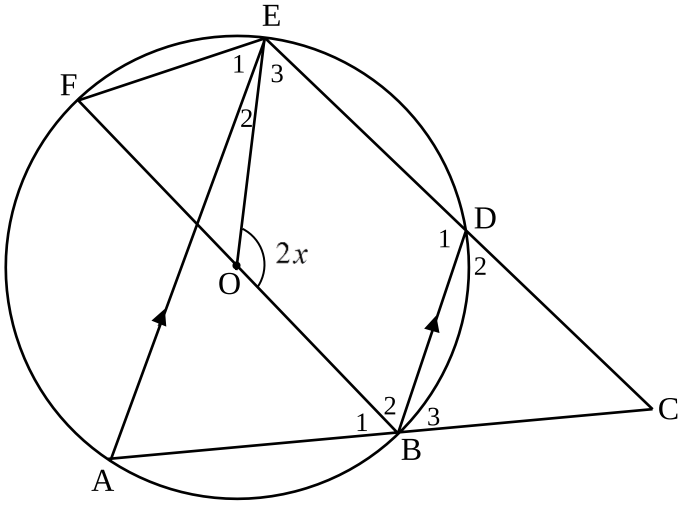
QUESTION 9

9.1 Use the diagram in your answer book to prove that $\frac{AD}{BD} = \frac{AE}{EC}$.



(5)

9.2 In the sketch below, O is the centre of the circle. A, B, D, E and F lie on the circumference. AB and ED are produced to meet at C. $AE \parallel BD$ and $E\hat{O}B = 2x$.



9.2.1 Determine, with reasons, each of the following angles in terms of x:

(a) $\hat{A}$ (1)

(b) $\hat{F}$ (1)

(c) $\hat{D}_2$ (1)

(d) $A\hat{B}D$ (1)

9.2.2 Prove that $BC = CD$. (2)

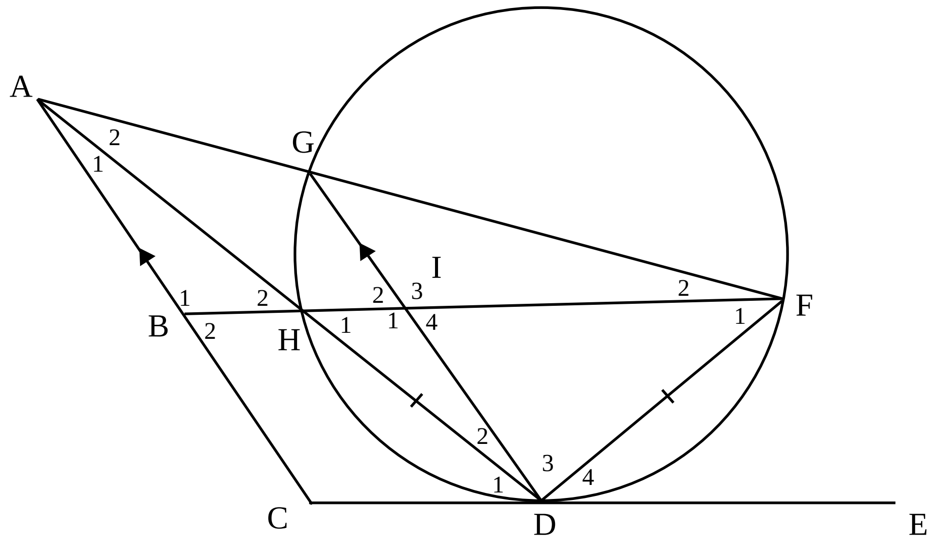
9.2.3 Prove that BCEO is a cyclic quadrilateral. (3)

9.2.4 It is further given that AB : AC = 3 : 5 and ED = 15 cm.
Calculate the length of DC. (3)

[17]

QUESTION 10

In the diagram below, $HD = DF$. G, H, D and F lie on the circumference of the circle. CE is a tangent to the circle at D. $AC \parallel GD$.



Prove that:

10.1 $\Delta ACD \,|||\, \Delta DIH$. (4)

10.2 $AB \cdot AD = AC \cdot AH$. (3)

10.3 AB is a tangent to circle AFH. (3)

10.4 $AB = \sqrt{BF \cdot BH}$. (5)

[15]